Fiscal Year:	FY 2016	**Task Last Updated:**	FY 02/03/2016
PI Name:	Wyrobek, Andrew Ph.D.		
Project Title:	Molecular Characterization of Choroid Plexus and Hippocampal Damage and Degenerative CNS Risks from Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	For immediate assistance please contact my administrator Caron LaMarsh clamarsh@lbl.gov, 510.486.5317		
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	04/04/2014	**End Date:**	04/03/2019
No. of Post Docs:	4	**No. of PhD Degrees:**	6
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	February 2016: Key members are: Sandhya Bhatnagar, Life Sciences Division, LBNL; Halina E. Witkowska, University of California, San Francisco; Bernard Rabin, University of Maryland, Baltimore Campus; Richard Britten, Eastern Virginia Medical School; Tore Straume, NASA Ames, Space Biosciences Division; Leif E. Peterson, Houston Methodist Research Institute. Changes in Personnel: Dr. Witkowska has formally retired. She is actively helping us to prepare manuscripts, and we are interviewing several laboratories to provide continued mass spectrometry analyses of this project.		
COI Name (Institution):	Peterson, Leif (The Methodist Hospital Research Institute) Straume, Tore (NASA Ames Research Center) Witkowska, Halina (University of California, San Francisco) Rabin, Bernard (University of Maryland, Baltimore Campus) Britten, Richard (Easterm Virginia Medical School) O'Banion, Kerry (University of Rochester)		
Grant/Contract No.:	NNJ14HP06I		
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Task Description:	A major goal of the NASA space radiation research program is to enable human exploration of space without exceeding limits for immediate and persistent risks to the central nervous system (CNS) from space radiation. Our research plan builds on the mounting evidence of the radiation-sensitivity of the hippocampus, which is important for memory and performance, and the choroid plexus (CP), which produces cerebral spinal fluid and maintains the blood-brain barrier. Using transcriptomic, proteomic, metabolomics, and immunohistochemical analyses of CNS tissue from rodents exposed to simulated space radiation, we have identified molecular and cellular changes associated with oxidative stress, neuro-inflammation, vasculature damage, changes in the protein composition of cerebral spinal fluid, and amyloid deposits such as those found in Alzheimer disease (AD). We propose new research to investigate the low-fluence responses and post-exposure-time dependence of these findings, to identify the molecular mechanisms of inter-individual variations in behavioral responses after radiation exposure, and to validate the predicted mechanisms using tissue of irradiated and behaviorally characterized rodents. Our specific aims are to: (1) Investigate the cellular and functional abnormalities in the hippocampus and CP of HZE-irradiated rats that are predicted by our multi-omics findings with the emphasis on defects associated with AD. (2) Investigate hippocampal and CP radiation damage responses associated with inter-individual variation in radiation-induced neurocognitive deficits, using proteomics, RNA expression, metabolomics, and in situ CNS analyses, with emphasis on tissue defects associated with AD. (3) Characterize the radiation sensitivity of persistent CP epithelial and endothelial damage in AD mouse models and examine their associations with neuro-inflammation and other defects associated with AD. This research is foundational to establish quantitative mechanism-based predictors for neuro-cognitive risks in individuals exposed to high-LET (linear energy transfer) space radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is designed to provide NASA with a mechanistic understanding of the risks for impaired neurocognitive performance and individual susceptibility for neurological diseases after exposure to high energy heavy ions (HZE) radiation, with emphasis on persistent neurological risks. Our approach: (a) employs state-of-the-art multi-omics and tissue technologies to investigate CNS damage response functions in animals exposed to low-fluence HZE particles (i.e., <1 hit per cell), (b) links the type and magnitude of molecular defects with neurocognitive performance results, and (c) validates the predicted damage response functions in brain tissue. Our project focuses on radiation damage to the hippocampus (site of adult neurogenesis; spatial and performance memories) and choroid plexus (CP, small cauliflower-like tissue in the ventricles adjacent to the hippocampus that produces cerebral spinal fluid (CSF) and maintains blood-brain homeostasis). Our research strategy distinguishes between two broad categories of risk mechanisms: (1) mechanisms related to exposure dose, and (2) inter-individual variations in radiation-induced CNS risks for neurocognitive performance and neurological diseases (i.e., Alzheimer Disease (AD). Our research is conducted in outbred strains of rats to capture the effects of genetic variation and in genetically defined double and triple mutation AD mice. Successful completion of this project will provide a mechanistic foundation for space-radiation-induced CNS risk that is based on quantitative molecular and cellular defects and their associations with measured neurocognitive and behavioral abnormalities. The project considers the time of onset and the persistence of effects, as well as inter-individual differences in responses. The project will employ space-relevant fluences of simulated HZE radiation, ensuring relevance for astronaut risk assessment. The results are expected to play a key role in helping to reduce the uncertainty associated with predicting the CNS risk for deep space exploration missions. This research is also designed to contribute to the identification of molecular targets for biological countermeasures to reduce CNS risks from space radiation.
	PROJECT OVERVIEW Our research goal is to understand the molecular mechanisms of individual variations in CNS tissue damage, behavioral performance, and risks for neurological diseases after exposure to HZE space radiation. Using the Sprague-Dawley rat model, animals are exposed to various types of HZE radation at the NASA Space Radiation Laboratory (NSRL), and tested for memory, operant behavior, movement, and anxiety. Tissues are collected for multi-omic profiling (proteomic, transcriptomic, and metabolomics) from cerebral spinal fluid (CSF), choroid plexus (CP), cortex, and hippocampus between 3 days to 22 months after exposure to generate molecular hypotheses for CNS damage and persistent neurological risks. These hypotheses are then tested under experimental protocol that allows for direct comparison of behavior performance, multi-omics predictions, and histopathological validation within the same animal. Genetic mouse models are also employed to validate the multi-omic and histological hypotheses related to CNS damage associated with risk for Alzheimer disease. PROTEOMICS AND TRANSCRIPTOMICS OF CSF, CP, CORTEX, AND HIPPOCAMPUS AS PREDICTORS OF PERSISTENT DAMAGE AFTER EXPOSURE HZE exposures as low as 10cGy 56Fe induced proteomic changes in the CSF that persisted to 2 months after exposure. The radiation-modulated CSF proteins are associated with various neurological and psychological diseases, reduced free radical scavenging, increased oxidative stress, vascular abnormalities, amyloid/fibril abnormalities, neuro-inflammation, and impaired glucose metabolism. Cortical and hippocampal transcript profiling at 22 months after exposures indicates that several of the predicted abnormalities in CNS function, including vascular abnormalities, amyloid accumulation, neuro-inflammation and impaired glucose metabolism may persist for life. The CP, which produces CSF, may be sensitive for persistent radiation damage to the CNS. CHOROID PLEXUS (CP) VASCULAR FIBROSIS AND HISTO-PATHOLOGY As predicted by our multi-omic findings, we inspected the CP blood vessels and epithelium from 4 to 9 months after 56Fe exposures for histological damage and persistent changes in protein expression. Beginning at ~4 months and increasing to 22 months after exposure, there was a progressive increase in CP endothelial vascular fibrosis, including partial and full occlusions composed of fibrinogen and other proteins. Also, we found increased morphometric distortions of the CP cuboidal epithelium at these time points, along with reduced epithelial expression of transthyrethin (TTR), a major CSF protein associated with normal beta-amyloid clearance. TTR protein expression was dramatically reduced in the epithelium of the CP, with detectable reductions in animals exposed to 10cGy. CNS PROTEIN AMYLOIDS AND AGGREGATES

Task Progress:	As predicted by our multi-omics findings, the frequencies of amyloids were increased throughout the CNS at 9 and 22 months after exposure (2-6 fold, age-matched). The same animals had increased frequencies of extracellular radially-diffused aggregations of beta-amyloid staining but without an apparent central amyloid. By image analyses we found increased beta-amyloid staining in the neuron-rich regions of the hippocampus and dentate gyrus. Irradiated animals also showed increased frequencies of extracellular TTR protein aggregates. The observation of TTR protein aggregates in CNS after radiation exposure is novel and TTR has been hypothesized to function in the scavenging of neurotoxic amyloid beta oligomers. PROTEOMICS OF INDIVIDUAL VARIATIONS IN ANXIETY SPATIAL MEMORY LEARNING IN SHAM AND IRRADIATED ANIMALS For the analyses of individual variation in radiation-induced anxiety, rats were exposed to 56Fe particles (group n=40), and assigned into the most or least anxious groups based on their performance on the elevated plus maze at ~2 months after exposure. By proteomic analyses, we found two categories of anxiety-associated proteins in the hippocampal and CP: those associated only with radiation exposure and those found in both irradiated and unirradiated animals. Using bioinformatics, the proteins that discriminated between anxiety states in both irradiated and unirradiated animals shared several upstream regulator genes known to be associated with the risk for Alzheimer Disease. The proteins that discriminated between anxiety states only in the irradiated animals, shared well-known upstream regulators of radiation damage response, such as TP53 and MYC. Anxiety discriminating proteins were associated with neuro-inflammation, neurotangle and amyloid biology, neurotransmitters, and synaptic activity. A confirmatory experiment with expanded behavior testing, multi-tissue multi-omics, and additional in situ endpoints is in progress. In a separate HZE exposure experiment, we reported that animals who are poor performers for spatial memory learning on the Barnes maze showed a linear dose-dependent decline in learning performance with an effect-doubling dose of ~15cGy. High-performance learners showed no detectable radiation effect at doses up to 20 cGy. WORKING MODEL Our working model is that the CP and CSF, which are major components of the blood-CSF brain barrier, are sensitive to radiation-induced abnormalities that persist for many months after exposure. The radiation-modulated changes in transcript and protein profiles predict a spectrum of late-onset CNS damage, such as endothelial fibrosis, protein aggregation and amyloids, impaired immune function, and abnormal glucose metabolism. Our model predicts that individual susceptibility to radiation-induced anxiety and memory/learning deficits are associated with the differential expression of CP and hippocampal proteins, and several candidate proteins are under investigation. Our approach is designed to provide a novel window toward understanding the important class of CNS mechanisms that predict individual variations in the magnitudes of risks for persistent CNS tissue damage and late-onset neurological diseases. This information may also allow us to predict individual susceptibility to space radiation exposures and may lead to the future development of mechanism-based molecular countermeasures.

Bibliography Type:	Description: (Last Updated: 06/19/2025)
Abstracts for Journals and Proceedings	Wyrobek AJ, Rabin B, Bhatnagar S, Albertolle M, Straume T, Witkowska E. "Simulated space radiation induces changes in CNS proteomes associated with deficits in neurocognitive performance." Invited Symposium Talk. 15th International Congress of Radiation Research, Kyoto, Japan, May 25-29, 2015. 15th International Congress of Radiation Research, Kyoto, Japan, May 25-29, 2015. , May-2015
Abstracts for Journals and Proceedings	Wyrobek AJ, Witkowska E, Bhatnagar S, Albertolle M, Straume T, Rabin B. "CNS expression changes predict choroid plexus histopathology and amyloid accumulation after cranial exposure to simulated space radiation." Invited Symposium presentation. Space Radiation and Heavy Ions in Therapy Symposium 2015 (SRHITS2015), Osaka, Japan, May 22-24, 2015. Space Radiation and Heavy Ions in Therapy Symposium 2015 (SRHITS2015), Osaka, Japan, May 22-24, 2015. , May-2015
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Abstracts for Journals and Proceedings	Wyrobek AJ. "Neurological risk and resilience of young adults exposed to environmental neuro-toxicants: molecular lessons from a rodent model." Invited Lecture Series. Aquatic Park Speaker Series by LBNL and QB3 EBIC, Berkeley, CA, September 2015. Proceedings of Aquatic Park Speaker Series by LBNL and QB3 EBIC, Berkeley, CA, September 2015. , Sep-2015
Abstracts for Journals and Proceedings	Wyrobek AJ. "Molecular Profiling of Neurocognitive Performance and Risk for Neurological Disease after Exposure to Ionizing Radiation." Invited Symposium Speaker in symposium, "Gene-Environment Interactions and Neurological Disorders." 46th Annual Meeting of the Environmental Mutagenesis and Genomics Society meeting, New Orleans, Louisiana, September 26-30, 2015. Environmental and Molecular Mutagenesis. 2015 Aug;56(Suppl 1):S1-91. Supplement: Environmental Mutagenesis and Genomics Society 46th Annual Meeting. http://dx.doi.org/10.1002/em.21969 , Sep-2015
Articles in Peer-reviewed Journals	Britten RA, Jewell JS, Miller VD, Davis LK, Hadley MM, Wyrobek AJ. "Impaired spatial memory performance in adult Wistar rats exposed to low (5 to 20 cGy) doses of 1 GeV/n 56Fe particles. " Radiation Research. 2016 Mar 1;185(3):332-7. https://doi.org/10.1667/RR14120.1 , Mar-2016
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